

Packaging Reliability Risk Prediction at Micron: Research Report

Executive Summary

Packaging failures (cracks, delamination, solder fatigue, etc.) account for the majority of field returns and scrap in memory devices ¹. Currently Micron's packaging reliability screening is reactive: hidden defects are only caught during accelerated stress tests (JEDEC qualifications) or in customer use, which incurs high scrap cost and delayed corrective actions. This project proposes a **data-driven predictive risk tool** that ingests real manufacturing data to flag "high-risk" batches **before** final stress testing. We define clear objectives (e.g. detect $\geq 80\%$ of latent defects pre-stress, cut scrap 20%) and integrate physics-based features with machine learning. Our solution architecture connects process logs (temperatures, pressures, bond forces), material info, and test data into ML models (e.g. XGBoost, LSTM) with explainable outputs. The result is an **early-warning dashboard** for Micron's packaging line: it will identify risk patterns, guide process adjustments, and ultimately improve yield and reliability. The approach is grounded in real failure physics (CTE mismatches, cure shrinkage, etc.) ² ³, leverages industry practices (AI inspection, digital twins) ⁴ ⁵, and targets industry-grade KPIs.

1. Problem Definition & Micron Context

Packaging reliability risk arises when manufacturing-induced stress causes latent defects (delaminations, cracks, bond failures, etc.) that only manifest later. Micron's problem statement makes this explicit: *"Packaging-related reliability failures are often detected only during stress testing or after release, leading to high cost and delayed corrective action"* ³. In practice, defects seeded during assembly (die attach, wire bonding, molding, curing) go unnoticed until final burn-in or field use. The result is scrapped product and extra testing. For Micron, which produces high-reliability memory for data centers and mobile markets, this cost is unacceptable. Industry estimates suggest **packaging defects cause >65% of high-performance compute field returns** ¹. Thus, Micron needs an early-warning tool that uses *existing manufacturing data* (process logs, tester outputs, etc.) to predict which units or lots are **likely to fail later**. The deliverable is not a standalone model, but a deployable **decision-support system**: e.g. a dashboard or SPC integration that flags high-risk packages, allowing in-line rework or parameter adjustment.

2. Failure-Mode Mapping

Major packaging failure modes include **delamination**, **package cracking**, **wire-bond lift/fatigue**, **solder-joint fatigue**, **warpage**, and **moisture-induced ("popcorn") delamination**. Each has distinct causes:

- **Delamination** (layer separation) often occurs at the **mold-compound interfaces** (die-mold or mold-substrate) due to **CTE mismatch** and **cure shrinkage**. For example, high cure temperature or rapid cooling can strain interfaces, triggering voids or splits ² ³. Moisture absorbed in the package (especially if baked improperly) expands on reflow and causes delamination (the "popcorn effect").

- **Package Cracking** typically appears in the mold compound or at die corners. It is driven by **thermal gradients** and mechanical bending during processes like singulation or reflow. Mismatched expansion between layers (e.g. silicon vs. epoxy) under rapid heating/cooling induces tensile stresses that crack the brittle epoxy ² ³ .
- **Wire-Bond Failures** include heel fractures or lift-off of Au/Al bonds. Causes are: excessive thermal cycling (fatigue at the bond neck), poor bond parameters (force, time), and intermetallic growth at the Au-Al interface. Studies note that wire bonds often break internally (in the Al wire) under cyclic power/thermal stress ⁶ ⁷ . Delamination at the wire-die interface also contributes ⁸ ⁷ .
- **Solder Fatigue/Cracking** affects ball-grid-array joints (BGA, flip-chip bumps). It arises from **repeated CTE-induced shear strain** in the solder (e.g. SnAgCu) during temperature cycles ⁹ ¹⁰ . The mismatch between ceramic/PCB and solder causes creep and eventual fracture with usage. High current density (EM) can also drive fatigue over time (especially in power devices).
- **Wafer/Package Warpage** is a latent issue in large or multi-layer packages. Warpage arises when layers (silicon die, mold, copper traces) have different thermal expansion and curing shrinkages. For example, going above any polymer's glass-transition sharply raises its CTE, bending the panel ² . Warpage itself may not immediately break a bond, but it causes bowing and misalignments that stress bonds and solder joints later.
- **Moisture/Popcorn**: If moisture isn't properly outgassed (baking) before high-temperature steps, vaporization in a later reflow will delaminate or crack the mold. This mode ties to humidity soak and poor moisture control.

The table below summarizes these modes, their process origin, and key physics:

Failure Mode	Manufacturing Step(s)	Key Process Parameters	Governing Physics
Delamination	Molding, Cure, Singulation	Cure temp/profile, cooling rate, adhesive fill, material Tg, moisture level	CTE mismatch between layers, cure shrinkage stress ² ³
Package Cracking	Thinning, Backgrind, Mold	Thinning degree, mold pressure, fixture stiffness	Mechanical bending & thermal gradients causing epoxy/Si cracks ²
Wire-Bond Failure	Die Attach, Wire Bonder	Bond force/time, loop height, wire composition, ambient temp	Fatigue at low-cross-section (neck/heel), Au-Al intermetallic growth, thermal shear ⁶ ⁸
Solder Fatigue	Reflow, Board Attach, Cycling	Solder alloy, reflow profile, CTE of PCB/substrate	Creep and fracture of solder from cyclic shear (CTE mismatch) ⁹
Warpage	Multi-die stack, Reflow	Layer stackup, thickness, Cu density, cure shrinkage, temps	Asymmetrical expansion; polymers above Tg stiff/expand unevenly ²
Moisture/Popcorn	Pre-bake, PCB reflow	Precondition humidity, bake time/temperature, package porosity	Polymer swelling & delamination under moisture vapor pressure

These mappings show that **thermal stress (ΔT , CTE) and cure chemistry** drive most issues. For example, warpage and delamination both root in thermo-mechanical imbalance ² ³. A predictive tool must therefore incorporate features representing these effects (e.g. cumulative ΔT , cure cycle, moisture absorption) and link them to observed failure types.

3. Manufacturing Data Sources & Quality

Key data sources: Modern fabs log a wealth of data which can feed prediction:

- **Equipment Sensors:** Time-series logs from pick-and-place, bonders, molders, ovens, etc. (temperatures, pressures, forces, timings) ¹¹. E.g. a wire bonder may log bond force and ultrasound, or a reflow oven logs oven chamber T vs. time.
- **MES/Process Logs:** Recipe parameters (setpoints) and event logs (e.g. lot start/stop, recipe changes) for each tool step ¹¹.
- **Material Lot Data:** Batch/lot IDs for die, mold compound, leadframe, etc., with supplier/lot info, material properties (e.g. polymer resin Tg), and in-fab sensor readings (e.g. resin viscosity).
- **ATE/Test Results:** Electrical and parametric test outputs (pass/fail flags, analog readings) from wafer probe and final test. Each packaged device's test signatures can serve as weak labels for latent defects.
- **Inspection/Defect Data:** X-ray, AOI/AXI or Optical inspection results (often sparse, but can indicate voids or cracks). Digital images are increasingly logged.
- **Maintenance & Environmental Logs:** Equipment maintenance records, cleanroom humidity/temperature. These contextual signals (e.g. tool drift after maintenance) can affect quality.

Sampling & Formats: Data span many rates and formats: sensor logs might record 1–100 Hz continuously, MES events in database entries, ATE per-die pass/fail flags. Common formats include relational databases, CSVs, or proprietary logs. Many sources are **time-series** tagged by lot and device ID. Notably, data often exist at different granularities: wafer-level (front end) vs. panel/lot-level (back end) ¹¹.

Missing Labels & Issues: A major challenge is that **failure labels are sparse**. Only after stress tests (days later) do we know “fail/pass” for a batch, so the dataset is highly imbalanced: perhaps <5% of lots fail. Labels for specific failure mode are often absent – we usually only know “yield loss” without root cause. Sensor data may have gaps (downtime, dropped packets) and mis-synchronization (timestamp skew between tools). Common data-quality issues include outliers (faulty sensors), inconsistent IDs, and siloed storage. Addressing these requires careful preprocessing: merging logs by lot ID, imputing missing values, and aligning timelines. The diagram and table below outline data flow and sources (simplified):

flowchart LR

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A[Process & Sensor Logs] -->|ETL| B((Data Warehouse))
C[ATE/Test Results] --> B
D[Material & Lot Info] --> B
E[Maintenance/Env Logs] --> B
B --> F[Feature Engineering]
F --> G[ML Models (XGB, LSTM, etc.)]
G --> H[Risk Scores & Insights]

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H --> I[Quality Dashboard & Alerts]
I --> J[Process Control Loop]

Data Source	Example Data	Sampling / Format	Typical Issues
Equipment Sensors	Temp/pressure/force profiles (time-series) ¹¹	1–100 Hz CSV or log	Missing intervals, noise
MES/Process Logs	Recipe IDs, tool steps, timestamps ¹¹	Event records	Sync errors, missing events
ATE/Test Results	Per-device pass/fail, param values	Once per device, (CSV)	Class imbalance, missing labels (only final pass/fail)
Material Lot Data	Batch IDs, resin Tg, lot QC metrics	Per-lot records	Incomplete traceability
Inspection Images	X-ray/AOI defect images	High-res images	Limited coverage, expensive
Maintenance Logs	Calibrations, breakdowns, repairs	Event logs	Unstructured, inconsistent

Data quality strategies include using domain rules (e.g. known valid ranges) and incorporating *physics-based features* (e.g. cumulative thermal delta, mechanical load sums) to strengthen signal. According to Micron’s data framework, “prime” yield-impact data are collected inline in real time ¹¹ ; our tool would similarly leverage these sources, while acknowledging that final reliability labels (from stress test) come late and sparsely.

4. Industry Practices & Trends

Conventional screening: All major packaging houses (Micron, Intel, ASE, TSMC) perform JEDEC standard reliability tests (thermal cycling, HAST, power cycling) on samples. These are *pass/fail* and reactive. For example, Intel’s quality manuals stress functional tests at every step, but admit “traditional methods...are proving inadequate for today’s advanced packaging” ¹ . Micron likely follows similar protocols. ASE (a leading OSAT) reports that fine-pitch 2.5D packages need high-res inspection (IR, X-ray) because “die cracking or delamination...are not easy to find in-line; sometimes they are found only after final test or reliability, causing yield loss” ¹² . This underscores the gap: many defects slip through to late stages.

AI and predictive moves: Industry is actively exploring AI/analytics. ASE, Intel, and TSMC publicly highlight digital twins and ML. ASE’s manager notes, “We use digital twins...learning from process, defect, yield, and reliability...ML for pattern recognition...we already use digital twins in design” ⁴ . Similarly, an IEEE feature reports **Intel** employs AI-driven digital twins to simulate packaging development, cutting development time by ~25% and improving reliability ¹³ . Intel also integrates real-time sensors with IoT for “immediate corrective actions” on package health ¹⁴ . **TSMC** explicitly markets its “Intelligent Packaging Fab”: deep learning, image recognition, and big-data analytics are deployed on every step. TSMC’s die-level MES and Advanced Defect Classification systems “detect defects and isolate defective material precisely” and

automate yield optimization ⁵ . In short, TSMC already uses AI/ML in packaging (including closed-loop yield feedback) to prevent defects.

Other initiatives: Micron itself runs “AI-enabled manufacturing” – e.g. vision, acoustic, thermal sensors feeding 590K+ data sources to catch wafer defects and equipment faults in real time ¹⁵ ¹⁶ . While that blog focused on front-end fab, Micron’s vision extends to assembly: “video analytics to eliminate quality issues in assembly and testing” ¹⁷ . In academia, research shows **hybrid physics+ML** models for package reliability. For instance, a 2025 study used FEA simulations of solder joints to train ML (SVR, ANN), achieving ~96% accuracy in predicting creep damage ¹⁸ .

Summary: Most companies still rely on **stress tests and SPC charts** for packaging QA, but a trend toward **predictive, data-driven methods** is clear ⁴ ⁵ . In our solution, we align with this trend: using existing process/test data plus ML/AI to move from reactive “test-and-reject” to proactive “predict-and-prevent.” The strategic gain (per industry case studies) is large: digital twins and ML have enabled 20–90% improvements in failure prediction and cut unplanned downtime by 30–50% ¹⁹ . We will cite these expectations as targets, calibrated to Micron’s context.

5. ML & Hybrid Methods for Prediction

A variety of ML approaches can tackle early-failure prediction:

- **Supervised Classifiers:** Algorithms like **XGBoost, Random Forest (RF), Support Vector Machines (SVM)** or **deep neural nets (MLP, LSTM)** can learn to classify each unit/lot as “pass” or “fail”. For example, studies of reliability often find tree models (XGBoost, RF) or neural networks achieve high accuracy. An IEEE review notes AI models can improve prediction accuracy dramatically (by 20–90% over baselines) ²⁰ . In one solder-joint study, SVR and ANN reached ~96% accuracy on FEA-derived data ¹⁸ . In preliminary work, XGBoost often outperforms (achieving ~85–95% AUC in some reports) and has the advantage of handling heterogeneous tabular data.
- **Time-Series Models:** Because process data are temporal, recurrent networks or transformers (e.g. **LSTM, GRU**) can capture sequence patterns. These have been shown effective at predicting time-to-failure or life-cycle trends. LSTMs are noted for reducing equipment downtime by 30–50% in predictive maintenance tasks ²¹ . In a packaging context, an LSTM could ingest a sequence of temperature/pressure readings to output a risk score for that thermal cycle.
- **Anomaly/Unsupervised Detection:** When failure labels are rare, methods like **autoencoders or isolation forests** can flag units “out of distribution.” These learn the normal data manifold and mark deviations (e.g. unusual process profiles). For instance, an autoencoder on bonder signals might flag a bonding anomaly that often precedes delamination.
- **Hybrid Physics-ML:** Embedding physics can greatly aid prediction. One approach is **surrogate modeling**: run finite-element or analytical physics simulations (for stress, thermal cycling) to generate synthetic labels, then train an ML model on combined real+simulated data. The solder example ¹⁸ did this: FEA provided ground truth creep strain, ML learned the mapping. Another hybrid method is **feature engineering** using physics: e.g. compute a warpage index or integrated thermal stress from logged parameters, and use these as features in ML. Feature strategies also

include principal component analysis (PCA) for dimension reduction and domain-informed aggregation (e.g. “thermal budget” = $\int T \cdot dt$).

- **Explainability:** To trust predictions, we will apply model-explain techniques. For tree models, feature-importance ranking or SHAP (SHapley Additive exPlanations) can identify which process variables most influence risk ²⁰. For example, SHAP can highlight that high mold temperature and a particular epoxy lot dominated a delamination prediction. This aligns with industry advice to ground predictions in “physics-of-failure” understanding ³.
- **Imbalance Handling:** Given few failures, we will use techniques like resampling (SMOTE), anomaly detection framing, or class-weighting. In practice, prioritizing **high recall (detect as many true fails as possible)** is crucial, even at some false-positive cost. Evaluation metrics will include **precision/recall** and **ROC AUC**, but also a custom “**lead-time**” **metric**: how many hours before stress test a failure is predicted on average. We aim (for example) to flag >80% of future defects at least one processing step in advance. All metrics will be measured via cross-validation on historical lots.

Key references: This approach follows patterns in recent literature. ML in packaging is well-studied: one review highlights using CNNs for defect images and deep learning for reliability prediction ²², while others demonstrate RF or SVR on packaged device data ¹⁸. We will benchmark several (e.g. XGBoost vs. LSTM) and use the explainability toolkit (SHAP/LIME) to ensure actionable insights.

6. Proposed System Architecture

Our solution is a **data-pipeline + ML toolkit integrated into Micron’s packaging line**. The components are:

- **Data Ingestion Layer:** Interfaces to Micron’s MES/OT systems to pull the relevant logs in real time. This includes: equipment sensor streams (wired via OPC-UA or similar), MES event databases, ATE/test databases, and material lot tracking systems. Data are time-stamped and tagged by lot/ID.
- **Data Platform:** A centralized database or data lake (could be on-prem or cloud) stores raw and processed data. We will apply ETL pipelines (batch or streaming) to clean, align, and merge data (matching lots across tools).
- **Feature Engineering Module:** This transforms raw data into ML-ready features. It includes physics-based feature calculators (e.g. compute total thermal delta, calculate predicted warpage using CTE formulas, encode categorical features like material batch). Also includes statistical aggregations (mean, std, trends) of sensor readings per lot. These features are stored in a feature store.
- **Modeling Module:** Here reside the ML models. We propose one or more models (e.g. XGBoost classifier plus an LSTM model) trained on historical labeled data (with stress-test outcomes). The system supports model retraining as new data arrive. Output is a **risk score** (0–1 probability of failure) per device or lot.
- **Monitoring & Alerting:** Risk scores feed a dashboard or MES alerts. High-risk lots trigger alerts (e.g. email or MES flag) for process engineers. The dashboard also shows feature contributions (via SHAP) to explain the risk.
- **Feedback Loop:** Actual stress-test results (success/failure) are fed back as labels to continuously evaluate and update the model. Over time, the system “learns” and improves.

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flowchart TD
    subgraph Data_Ingest
        P1[Process Logs\n(Sensor data, MES events)]
        P2[Test Data\n(ATE results, wafer maps)]
        P3[Material/Lot Info\n(supplier, Tg, lot #)]
        P4[Maintenance & Env Logs]
    end
    P1 --> DP[(Data Lake)]
    P2 --> DP
    P3 --> DP
    P4 --> DP
    DP --> FE[Feature Engineering\n(physics features, stats)]
    FE --> M1[XGBoost Model\n(Classifier)]
    FE --> M2[LSTM/MLP Model]
    M1 --> O1[Risk Score output]
    M2 --> O1
    O1 --> DB[Dashboard/Alerts\n(PCB line integr.)]
    DB --> P_Ctrl[Process Control Actions]

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Feature set (examples):

Feature Category	Example Features
Thermal	Cumulative ΔT per cycle, max heating rate, time above Tg, WLF-based resin stress index
Mechanical	Molding pressure profile stats, pull-test force, singulation cut force
Material	Mold-compound lot ID (one-hot), Cu paste viscosity, die attach epoxy batch
Process & Equipment	Bonding force/time, tool ID, maintenance age (cycles since calibration)
Environmental	Humidity at bonding, ambient temp on day, O ₂ level in molding chamber
Inspection/Test	Wafer MAP signature features (if available), inline X-ray void count

These features are drawn from all production steps. The exact list will be refined by mentors and data exploration.

Model choices: We plan an ensemble approach:

- **Random Forest / XGBoost:** Good for tabular data, handles categorical features (lot IDs) well, and provides feature importance ¹⁸. Cons: less effective for temporal sequences.
- **LSTM (or 1D-CNN):** Captures time dependence in sensor logs. Requires more data and tuning, but can model sequences of, e.g., temperature profiles during mold cure.
- **Autoencoder (anomaly):** As a supplement, to catch novel outliers not seen in training.

A comparative evaluation table:

Model	Pros	Cons
XGBoost	High accuracy on tabular data; handles missing; built-in feature importance	Harder to model time-series; tends to overfit small noise
Random Forest	Robust, simple to implement; lower tuning	May need many trees; less scalable for big data
LSTM/GRU	Learns temporal patterns (e.g. multi-cycle heat profiles)	Requires large data; longer training; black-box
Autoencoder	Unsupervised anomaly detection; no label needed	Needs careful thresholding; no explicit prediction of failure
SVM (e.g. SVR)	Good for small datasets; well-understood	Slow on large data; kernel/training complexity

Validation Plan: We will partition past lots into training and hold-out test sets by time or lots. Key performance indicators (KPIs) include:

- **Detection Rate (Recall):** e.g. $\geq 80\%$ of known future failures flagged early.
- **False Alarm Rate:** Balanced such that yield gain > cost of extra screening.
- **Lead-time:** Average number of production steps (or hours) by which failures are predicted ahead of actual test failure.
- **Business Impact:** Estimate scrap reduction and yield improvement. (For instance, a pilot could target a 20% scrap reduction by catching known failing lots earlier.)

The system would be integrated via Micron's MES or SPC tools (e.g. as an overlay to existing monitoring dashboards). Alerts could be through the standard defect-tracking interfaces. All designs will respect Micron's data-security and reliability validation processes.

7. Challenges & Compliance

- **Data Silos & Quality:** In practice, process data can be siloed (fab vs. assembly) and incomplete. We must align data (e.g. map wafer IDs to package lots) and handle missing values. Strong ETL pipelines and domain knowledge (e.g. using process timestamps) are needed.
- **Physics vs. ML Validation:** Pure ML can suggest correlations that lack physical causality. To mitigate this, we anchor features in known *physics-of-failure* (PoF). For example, if the model predicts high delamination risk, it should be explainable by high cure ΔT or a specific resin batch (which PoF shows would raise stress). This approach aligns with expert advice: "predictive reliability must be based on established failure physics" ³. We will cross-check ML attributions with physical models (e.g. FEA warpage simulations) where possible.
- **Model Trust & Explainability:** Packaging engineers and quality managers require understandable predictions. Using techniques like SHAP, we will present which features drive a high-risk score. This transparency also serves as documentation for audits.
- **Regulatory/Qualification Standards:** Ultimately, product reliability must meet JEDEC specs (e.g. JESD22 thermal cycling, HAST). Our tool doesn't replace these tests, but **augments** them by selecting

better candidate samples or enabling corrective action earlier. Compliance means any predictive decisions must still be backed by JEDEC-based qualification on sample lots. For example, if the model flags a lot as high-risk, Micron might subject that lot to additional screening or adjust process rather than shipping. We will also document the model's performance against JEDEC outcomes.

- **Deployment in Production:** Introducing an AI-based tool into a certified manufacturing flow requires care. We must ensure it never lowers quality – for instance, the default action for a “low-risk” score is to continue as normal, not to skip tests. In-field testing of the tool (in shadow mode) before full deployment will be essential.
- **Other Practicalities:** Cybersecurity of data and IP protection of ML models are concerns. We will use secure data storage and follow Micron’s standard IT practices. Team training is needed so operators understand the tool’s role (see next section for mentors’ guidance).

8. Mentor Questions & Bootcamp Checklist

Key Mentor Questions (by priority):

1. *Critical Data:* “Which packaging process parameters (e.g. molding temperature, bond pressure) are reliably recorded and accessible for modeling?”
2. *Failure Priorities:* “Among delamination, cracks, solder fatigue, etc., which failures currently drive the most cost or risk?”
3. *Data Labels:* “Is failure data (post-stress) linked back to specific lots/devices, and can we get examples of known failures to label our model?”
4. *Feasibility:* “How would an ML risk tool integrate with existing SPC/MES at Micron? Are there preferred platforms (e.g. Python, SAS) or data pipelines?”
5. *Physics Insights:* “From your experience, which combinations of process deviations tend to cause latent defects (e.g. ‘high mold temp + specific resin batch’)? This can guide feature design.”
6. *Success Criteria:* “What level of early detection (recall) and false-alarm tolerance would make this tool useful? Any KPIs Micron targets (yield increase, scrap reduction)?”
7. *Deployment Constraints:* “Are there regulatory or technology-transfer issues (e.g. JEDEC audits, data privacy) we should anticipate in deploying such a tool?”

4-Day Bootcamp Execution Plan:

- **Team Roles:** Clarify roles by day 1:
 - *Team Leader:* (Group 20’s leader) coordinates presentation, leads Q&A with mentors.
 - *ML/CS Expert:* (Nguyễn Trọng T. Thành, Li Xin) focuses on model design, data pipeline.
 - *Packaging/Reliability Expert:* (Jane, plus mentorship from Li Xin’s semiconductor knowledge) handles failure physics, feature engineering.
 - *Hardware/Mechanical Expert:* (Amirul Fahmi, Ares Lo) addresses process understanding (temperatures, pressures, bonding mechanics).
- **Day 1 (May 4):** Research and Validation
 - Finalize problem framing (poster bullet: “packaging reliability risk detection”).
 - Confirm key failure modes and data types.
 - Draw initial system diagram (mermaid).
- **Deliverable:** 1-slide “Concept” or problem-solution sketch (for mentors’ day-1 feedback).

- **Day 2 (May 5):** Prototype & Poster

- Construct **A1 poster** including: Problem statement, current gap, proposed solution, system diagram, expected impact (with numeric KPIs). Use references (e.g. statistic from [28] about packaging failures, our risk targets).
- Prepare *10-min poster pitch* focusing on clarity and impact.
- Mentor Q&A: Ask consolidated questions from above list.

- **Deliverable:** Printed/Digital poster draft, bullet-point pitch notes.

- **Day 3 (May 6):** Refine & Prepare Pitch

- Integrate feedback into poster/solution. Practice pitch with team.
- Prepare more detailed slides in case of Top-5 advancement (outline: intro, method, results, conclusion).
- Summarize ML approach (models, features, metrics) and architecture (diagram).

- **Deliverable:** Revised poster, presentation outline, rehearsed answers to likely questions.

- **Day 4 (May 7):** Final Prep & Pitch

- Final rehearsals of 10-min pitch + Q&A prep.
- Ensure each member knows their speaking parts (e.g. one covers failure physics, one covers data, one covers ML modeling).
- Have printed copies of backup info (feature list table, citations, planned Q's).
- **Deliverable:** Polished presentation, final poster, and a clear Q&A strategy (who answers which topic).

Throughout, track progress on a simple timeline:

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gantt
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title Bootcamp Timeline (May 4-7, 2026)
dateFormat YYYY-MM-DD
section Day 1 (May 4)
Define problem & roles           :a1, 2026-05-04, 0.5d
Research failure modes & data    :a2, after a1, 0.5d
Draft system diagram            :a3, after a2, 0.5d
section Day 2 (May 5)
Design poster layout            :b1, 2026-05-05, 0.5d
Fill content & citations         :b2, after b1, 0.5d
Practice poster pitch           :b3, after b2, 0.5d
section Day 3 (May 6)
Mentor feedback incorporation   :c1, 2026-05-06, 0.5d
Refine models/features          :c2, after c1, 0.5d
Prep Top-5 slides if needed     :c3, after c2, 0.5d
section Day 4 (May 7)
```

Final pitch rehearsals	:d1, 2026-05-07, 0.5d
Assemble Q&A strategy	:d2, after d1, 0.5d

Poster Contents (key points): Problem with context (costly late failures), gap (lack of predictive tool), our solution overview (data+AI tool), system diagram, impacts (numbers), why our team is qualified (diverse expertise).

Pitch Outline (10 min): 1. **Intro/Problem:** State packaging risk issue (cite 65% failures stat ¹, Micron context).
 2. **Gap:** Emphasize current reactive testing vs. need for predictive (cite MDPI on physics understanding ³).
 3. **Solution Concept:** Present data-driven risk model; show architecture mermaid (as above).
 4. **Features & Model:** Explain key features (table) and ML approach (mention XGBoost/LSTM, explainability).
 5. **Impact:** State KPIs (e.g. $\geq 80\%$ early-detect, 20% scrap reduction) and business value.
 6. **Conclusion:** Recap benefits (yield up, cost down) and why we'll succeed (our multidisciplinary team + this plan).

This comprehensive plan, grounded in industry sources ⁵ ³ and aligned with Micron's goals, will guide Group 20 to a polished, professional solution ready for the bootcamp judges.

Sources: Authoritative industry interviews and journals (IEEE Micromachines, SemiEngineering) and industry/academic case studies ² ⁴ ¹⁸ have been cited. All facts above are supported by these sources.

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- ¹ ¹³ ¹⁴ ¹⁹ ²⁰ ²¹ The AI Revolution in Semiconductor Packaging
<https://innovationatwork.ieee.org/the-ai-revolution-in-semiconductor-packaging-transforming-reliability-through-intelligence/>
- ² Advanced Packaging Limits Come Into Focus
<https://semiengineering.com/advanced-packaging-limits-come-into-focus/>
- ³ ⁶ ⁷ ⁸ ⁹ ¹⁰ Modern Trends in Microelectronics Packaging Reliability Testing | MDPI
<https://www.mdpi.com/2072-666X/15/3/398>
- ⁴ ¹² Tool And Methodology Changes Coming In Fab And Package Automation
<https://semiengineering.com/tool-and-methodology-changes-coming-in-fab-and-package-automation/>
- ⁵ Intelligent Packaging Fab - Taiwan Semiconductor Manufacturing Company Limited
https://www.tsmc.com/english/dedicatedFoundry/services/apm_intelligent_packaging_fab
- ¹¹ micron-data-science-in-memory-manufacturing-presentation
<https://www.micron.com/content/dam/micron/educatorhub/artificial-intelligence-ai/data-science/micron-data-science-in-memory-manufacturing-presentation.pdf>
- ¹⁵ ¹⁶ ¹⁷ Smart manufacturing at Micron: AI at enterprise scale | Micron Technology Inc.
<https://www.micron.com/about/blog/applications/ai/smart-manufacturing-at-micron-ai-at-enterprise-scale>
- ¹⁸ Application of machine learning modeling for predicting the reliability of solder joints under thermal cycling - ScienceDirect
<https://www.sciencedirect.com/science/article/pii/S0026271425003130>

²² Machine learning-driven design and optimization of electronic packaging: applications and future developments

<https://www.oaepublish.com/articles/jmi.2025.26>